

Kyle Fairbanks

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Portfolio: <https://kyle-fairbanks.github.io>

Mechanical Engineer with hands-on quality engineering experience supporting electromechanical manufacturing, troubleshooting production issues, developing test systems and fixtures, and validating products. Experienced in SolidWorks, prototyping, electrical and mechanical testing, root-cause analysis, and cross-functional manufacturing support.

Education

University of Arizona – Tucson, AZ

- B.S. Mechanical Engineering; Minor: Aerospace Engineering — Graduated May 2026
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Experience

ASSA ABLOY – Quality Engineer (June 2026 - Present)

- Support production of electromechanical access-control products through finished-goods audits, first-article inspections, electrical and mechanical testing, and manufacturing troubleshooting
- Work directly with production, quality, and engineering personnel to identify failure mechanisms, evaluate manufacturing issues, and support corrective actions and process improvements
- Investigate product and process nonconformances through functional testing, electrical diagnostics, dimensional inspection, and comparison against engineering specifications
- Investigate DMRs and perform root cause and corrective action (RCCA) for manufacturing and product issues

ASSA ABLOY – Quality Engineering Intern (June 2024 - June 2026)

LSP Snap-In DIN Rail Mounting Bracket

- Designed a 3D-printed snap-fit mounting bracket in SolidWorks that secures a power-supply board within its enclosure and mounts the bracket-board subassembly directly to DIN rail
- Iterated the design through geometry refinement, material selection, snap-fit performance, and UL compliance requirements

ControlID Hub Mounting Bracket

- Designed a 3D-printable press-fit mounting bracket in SolidWorks, balancing fit, wire routing, removability, and printability

Operator Guidance and Verification System

- Designed a multi-station operator guidance and verification system using LabJack I/O, lasers, LEDs, and switches for real-time assembly validation
- Developed a self-contained Python/Flask control interface with configurable test modes, logging, error handling, and hardware-in-the-loop validation

Multi-Door Automated Cycle Tester

- Delivered an automated cycling tester enabling 12 parallel door-cycle tests in the footprint of one existing rig
 - Integrated hydraulic, electronic, and mechanical subsystems in a modular architecture supporting automated cycling, monitoring, maintenance, and future expansion
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Projects

Dynamically Scaled Modular UAV — Senior Design Capstone, University of Arizona

- Designed a composite wing for a 1.38 m span UAV targeting 80 g mass under a 5G load requirement
 - Built a custom Python structural analysis tool using Classical Laminate Theory, Bredt-Batho torsion, and Tsai-Wu failure criteria to analyze the wing under load; design confirmed by a physical 5G wing loading test
 - Selected composite materials, carbon-fiber tube interfaces, and bonding agents for a lightweight UAV wing
 - Performed composite hand layup, vacuum bagging, bonding, and structural validation for the critical wing build
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Technical Skills

Manufacturing / Quality: Manufacturing troubleshooting, root-cause analysis, RCCA, first-article inspection, functional testing, product validation, continuous improvement, dimensional inspection

CAD / Enterprise Software: SolidWorks, Siemens Teamcenter, JD Edwards EnterpriseOne, Microsoft Office

Analysis / Testing: Structural/composite analysis, system verification, hardware validation, test fixture development

Prototyping: FDM 3D printing, MSLA 3D printing, DfAM, composite hand layup, vacuum bagging, hand fabrication

Materials: Composites, carbon-fiber tubes, carbon-fiber and fiberglass fabrics, 3D-printing polymers, adhesive bonding